



NeuroSpace

Complete evaluation export

Job ID	5b744921-dec2-46e1-88c4-806de5426538
Status	completed
Result timestamp	2026-05-11T19:15:36.868464
PDF generated	2026-05-11 20:21:38

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	mi
feature_selector	mi
qrnn_backend	qiskit
noise_enabled	True
noise_level	0.0100
mitigation_enabled	True
mitigation_runs	3
manual_overrides.rnn_epochs	30
manual_overrides.qrnn_epochs	20
manual_overrides.qrnn_batch_size	16

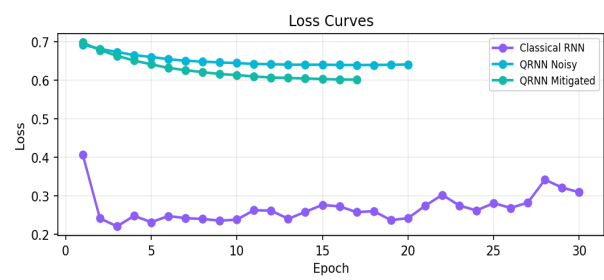
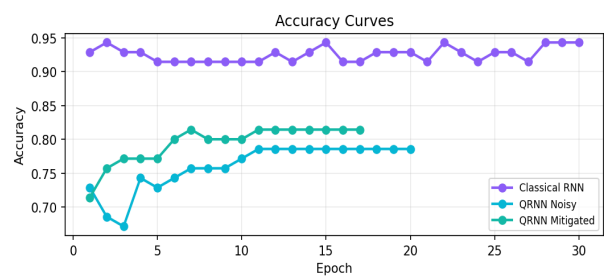
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	94.29%	88.57%	5.71%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	94.29%	94.29%	94.29%	94.29%	0.3085
QRNN Noisy	81.43%	82.29%	81.43%	80.11%	0.6410
QRNN Mitigated	88.57%	88.92%	88.57%	88.20%	0.6014

Learning Curves



Classical RNN

Metric	Value
Accuracy	94.29%
Precision	94.29%
Recall	94.29%
F1 Score	94.29%
Loss	0.3085
Train Loss	0.0518
Val Loss	0.3085
Train Accuracy	97.75%
Val Accuracy	94.29%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	4	44

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	95.65%	95.65%	95.65%	92.0000
1	91.67%	91.67%	91.67%	48.0000
macro avg	93.66%	93.66%	93.66%	140.0000
weighted avg	94.29%	94.29%	94.29%	140.0000
Accuracy			94.29%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.4590	0.4065	79.96%	92.86%
2	0.1851	0.2409	95.09%	94.29%
3	0.1525	0.2197	96.52%	92.86%
4	0.1342	0.2473	95.91%	92.86%
5	0.1191	0.2306	97.34%	91.43%
6	0.1437	0.2467	96.73%	91.43%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.1080	0.2411	96.73%	91.43%
8	0.0920	0.2392	97.55%	91.43%
9	0.0975	0.2345	97.14%	91.43%
10	0.0983	0.2374	97.75%	91.43%
11	0.0979	0.2616	96.93%	91.43%
12	0.0957	0.2609	97.34%	92.86%
13	0.1124	0.2393	96.93%	91.43%
14	0.0656	0.2576	98.77%	92.86%
15	0.0948	0.2756	97.34%	94.29%
16	0.0954	0.2717	96.93%	91.43%
17	0.0893	0.2567	97.96%	91.43%
18	0.0994	0.2594	97.14%	92.86%
19	0.0714	0.2360	97.75%	92.86%
20	0.0746	0.2412	97.34%	92.86%
21	0.0740	0.2733	97.14%	91.43%
22	0.0656	0.3018	98.16%	94.29%
23	0.0613	0.2739	98.16%	92.86%
24	0.0902	0.2611	97.34%	91.43%
25	0.0667	0.2802	97.55%	92.86%
26	0.0656	0.2679	97.75%	92.86%
27	0.0645	0.2816	97.75%	91.43%
28	0.0869	0.3414	97.75%	94.29%
29	0.0738	0.3207	97.34%	94.29%
30	0.0518	0.3085	97.75%	94.29%

QRNN Noisy

Metric	Value
Accuracy	81.43%
Precision	82.29%
Recall	81.43%
F1 Score	80.11%
Loss	0.6410
Train Loss	0.5549
Val Loss	0.6410
Train Accuracy	85.69%
Val Accuracy	78.57%
Train Size	489
Val Size	70
Test Size	140

QRNN Noisy Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	22	26

QRNN Noisy Classification Report

Class	Precision	Recall	F1 Score	Support
0	80.00%	95.65%	87.13%	92.0000
1	86.67%	54.17%	66.67%	48.0000
macro avg	83.33%	74.91%	76.90%	140.0000
weighted avg	82.29%	81.43%	80.11%	140.0000
Accuracy			81.43%	

QRNN Noisy Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6740	0.6925	76.69%	72.86%
2	0.6559	0.6814	78.32%	68.57%
3	0.6428	0.6734	80.16%	67.14%
4	0.6300	0.6652	82.21%	74.29%
5	0.6195	0.6603	83.03%	72.86%
6	0.6121	0.6546	83.23%	74.29%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.6026	0.6506	83.84%	75.71%
8	0.5960	0.6484	83.84%	75.71%
9	0.5890	0.6462	84.87%	75.71%
10	0.5861	0.6444	85.28%	77.14%
11	0.5825	0.6423	85.28%	78.57%
12	0.5761	0.6416	85.89%	78.57%
13	0.5725	0.6405	85.89%	78.57%
14	0.5688	0.6402	85.89%	78.57%
15	0.5670	0.6403	85.48%	78.57%
16	0.5651	0.6396	85.69%	78.57%
17	0.5589	0.6395	85.89%	78.57%
18	0.5592	0.6396	85.89%	78.57%
19	0.5597	0.6401	85.69%	78.57%
20	0.5549	0.6410	85.69%	78.57%

QRNN Mitigated

Metric	Value
Accuracy	88.57%
Precision	88.92%
Recall	88.57%
F1 Score	88.20%
Loss	0.6014
Train Loss	0.5092
Val Loss	0.6014
Train Accuracy	88.75%
Val Accuracy	81.43%
Train Size	489
Val Size	70
Test Size	140

QRNN Mitigated Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	89	3
Class 1	13	35

QRNN Mitigated Classification Report

Class	Precision	Recall	F1 Score	Support
0	87.25%	96.74%	91.75%	92.0000
1	92.11%	72.92%	81.40%	48.0000
macro avg	89.68%	84.83%	86.57%	140.0000
weighted avg	88.92%	88.57%	88.20%	140.0000
Accuracy			88.57%	

QRNN Mitigated Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6519	0.6981	81.80%	71.43%
2	0.6275	0.6781	84.25%	75.71%
3	0.6143	0.6635	84.87%	77.14%
4	0.5971	0.6510	85.48%	77.14%
5	0.5826	0.6412	85.69%	77.14%
6	0.5709	0.6317	87.12%	80.00%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.5636	0.6262	88.14%	81.43%
8	0.5548	0.6207	88.14%	80.00%
9	0.5478	0.6161	87.93%	80.00%
10	0.5376	0.6133	88.14%	80.00%
11	0.5354	0.6096	88.14%	81.43%
12	0.5308	0.6070	88.55%	81.43%
13	0.5268	0.6058	88.75%	81.43%
14	0.5177	0.6045	88.96%	81.43%
15	0.5168	0.6029	88.34%	81.43%
16	0.5159	0.6019	88.75%	81.43%
17	0.5092	0.6014	88.75%	81.43%